Fiscal Year:	FY 2016	**Task Last Updated:**	FY 05/09/2016
PI Name:	Lee, Stuart M.C. Ph.D.		
Project Title:	Metabolomic and Genomic Markers of Atherosclerosis as Related to Oxidative Stress, Inflammation, and Vascular Function in Twin Astronauts		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77058-2749	**Congressional District:**	36
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2013-14 HERO NNJ13ZSA002N-TWINS
Start Date:	07/15/2014	**End Date:**	12/31/2018
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Grant number change in late 2017 to 80NSSC18K0037 and end date change to 12/31/2018 per NSSC information (Ed., 1/9/19) NOTE: Extended to 9/30/2018 per PI (Ed., 5/7/18) NOTE: Extended to 4/18/2018 per NSSC information (Ed., 8/25/17)		
Key Personnel Changes/Previous PI:	No changes		
COI Name (Institution):	Hargens, Alan Ph.D. (University of California, San Diego) Macias, Brandon Ph.D. (University of California, San Diego) Rana, Brinda Ph.D. (University of California, San Diego) Sears, Dorothy Ph.D. (University of California, San Diego) Smith, Scott Ph.D. (NASA Johnson Space Center) Stenger, Michael Ph.D. (Wyle Laboratories, Inc.)		
Grant/Contract No.:	80NSSC18K0037 ; NNX14AN75G		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	Astronauts participating in long-duration missions may be at an increased risk of oxidative stress and inflammatory damage due to radiation, psychological stress, altered physical activity, nutritional insufficiency, and hyperoxia during extravehicular activity. This proposal extends a current NASA-funded investigation to determine whether these spaceflight factors contribute to an accelerated progression of atherosclerosis. We propose to measure blood and urine biomarkers of oxidative stress and inflammation as well as arterial structure and function (carotid intima-medial thickness and brachial artery flow-mediated dilation) in one twin astronaut before, during, and after long-duration spaceflight and in his twin serving as a ground-based control. Furthermore, we will measure metabolomic and genomic (DNA methylation, mRNA gene expression, telomere length) markers to elucidate the molecular mechanisms involved. We predict that the space-flown twin will experience elevated biomarkers of oxidative stress and inflammatory damage, altered arterial structure and function, accelerated telomere shortening, dysregulation of genes associated with oxidative stress and inflammation, and a metabolic profile shift that is associated with elevated atherosclerosis risk factors. Conversely, these will not be observed in the ground-based twin. This study of twins affords a unique opportunity to examine the spaceflight-related atherosclerosis risk independent of the confounding factors of different genotypes.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work seeks to improve understand of the potential cardiovascular disease risk among long-duration space flown astronauts and biomarkers of atherosclerotic risk in normal Earth-bound populations.
Task Progress:	Pre-, in-, and post-flight biological samples and ultrasound images for both the space-flown and ground twin currently were collected. Landing day for the flight twin was for March 1, 2016. We have optimized our mitochondrial flux assays on a twin cohort undergoing simulated microgravity (head-down tilt bed rest, HDBR) in which one twin was participated in a lower body negative pressure exercise countermeasure which has been shown to prevent many physiological dysfunctions related to bed rest. Comparisons of the basal and maximal respiration between the discordant co-twins at bed rest day 5 suggest that there may be a plasma factor (e.g., metabolite) that increases respiration in the exercise twin. Using this approach demonstrated in these bed rest subjects, we also will investigate the twin astronaut pair discordant for space travel.
Bibliography Type:	Description: (Last Updated: 11/18/2025)
Abstracts for Journals and Proceedings	Lee SMC, Rana BK, Stenger MB, Sears DD, Smith SM, Macias BR, Hargens AR, Sharma K, De Vivo I. "Metabolomic and genomic markers of atherosclerosis as related to oxidative stress, inflammation, and vascular function in twin astronauts." 2016 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 8-11, 2016. 2016 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 8-11, 2016. , Feb-2016
Abstracts for Journals and Proceedings	Rana BK, Siamwala J, Patel H, Macias BR, Lee SMC, Stenger MB, Hargens AR. "Twins In Space: MZ Twins Discordant for Space Travel." 46th Behavioral Genetics Association Annual Meeting, Brisbane, Australia, June 21-23, 2016. 46th Behavioral Genetics Association Annual Meeting, Brisbane, Australia, June 21-23, 2016. , Jun-2016
Significant Media Coverage	Lee SMC with Lori Meggs (NASA Public Affairs). "Space Station Live: The Heart of the Matter. Video interview." Video interview, July 8, 2015. https://www.youtube.com/watch?v=GQsD4qmwhzk , Jul-2015